Annals of Communications in Mathematics
Volume 9, Number 1 (2026), 1
DOI: 10.62072/acm.2026.0900X
ISSN: 2582-0818
https://www.technoskypub.com/journal/acm/

Research article

A class of A-stable two-step multi-derivative hybrid block methods for stiff initial value problems

O. V. Atabo[1] *, E. O. Senewo[2], B. D. Michael[3] I. G. Ezugorie[4] Q. O. Ahman[5] B. C. Agbata[6]

[1,2,3,5,6]Department of Mathematics and Statistics, Confluence University of Science and Technology, Osara, Kogi State, Nigeria.
[4]Department of Mathematics and Statistics, Confluence University of Science and Technology, Osara, Kogi State, Nigeria.

*Corresponding Author: O. V. Atabo. Email: ataboov@custech.edu.ng

Received: 00; Accepted: 00; Published: 00

Citation: O. V. Atabo, E. O. Senewo, B. D. Michael, I. G. Ezugorie, Q. O. Ahman, B. C. Agbata, A class of A-stable two-step multi-derivative hybrid block methods for stiff initial value problems. Ann. Commun. Math. 9 **(2026)**, 1. https://doi.org/10.62072/acm.2026.0900X

Abstract: The simultaneous approximation of solutions at discretized mesh points remains a central theme in numerical analysis, particularly in the numerical treatment of differential systems via block solution approaches. In this study, a class of two-step hybrid block methods featuring multi-derivative functions, dynamically collocated for enhanced stability through balanced discretization constants, is proposed for solving initial-value problems in ordinary differential equations. The formulas are systematically derived through interpolation and collocation techniques utilizing polynomial power series expansions. Owing to their structural design, the proposed methods operate as fourth-derivative class techniques. Although they require the evaluation of higher-order derivative functions, they rigorously maintain A-stability and ensure high-order convergence, as established through comprehensive theoretical analysis. Numerical experiments conducted on a diverse suite of stiff and non-stiff initial-value problems confirm their robustness, demonstrating superior accuracy, computational efficiency, and convergence compared with state-of-the-art existing approaches.

Mathematics Subject Classification: 65L05, 65L20
Keywords: Multi-derivative methods, Block integration, A-stability, Convergence analysis, Stiff systems.

1 Introduction

Equations of the form given in Eq. (1) arise frequently in diverse scientific domains including biology, physics, and engineering. These mathematical models are generally intractable analytically, necessitating the development of robust and stable numerical integrators [1]. As articulated by [2], the essential prerequisites for numerical convergence are the synthesis of adequate stability with a sufficiently high order of accuracy. However, the simultaneous enforcement of high order and broad stability regions imposes strict theoretical

Copyright © 202X The Author(s). Licensee Techno Sky Publications.
This article is an open-access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

limitations on classical Linear Multi-step Methods (LMMs), particularly the Dahlquist barrier which asserts that the maximum order of an unconditionally stable LMM cannot exceed $p = 2$ [3].

To circumvent these inherent restrictions, two primary strategies have traditionally been employed: the utilization of block backward differentiation formulas (BBDFs) and the incorporation of intra-step (off-grid) collocation points into LMMs. While both paradigms enhance the stability domain and convergence order, intra-step formulations can introduce algorithmic complexities and increase computational overhead due to the evaluation of functions at off-grid points. Hybrid block methods, which occasionally rely on predictor–corrector implementations, also risk increased structural complexity. Over recent decades, these limitations have been mitigated by formulating LMMs designed for direct block implementations (see [4–10,12–15]).

An ordinary differential equation is classified as stiff when its Jacobian matrix exhibits eigenvalues with large negative real parts, forcing conventional numerical methods to employ prohibitively small step sizes to maintain absolute stability. Consequently, numerous studies have optimized stiff solvers using BBDFs, generalized Adams-type approaches, or modified continuous extended LMMs that preserve accuracy while accommodating stiffness [16–22]. For instance, [24] established a class of continuous extended LMMs with step numbers $k \leq 11$, achieving A-stability up to a theoretical maximum order of $p = k + 3$, and verifying enhanced accuracy across linear and nonlinear stiff initial-value problems (IVPs). Similarly, [25] formulated generalized Cash-type second-derivative extended backward differentiation formulas (EBDFs) via boundary-value frameworks. Their schemes achieved A-stability for all $k \geq 1$ with order $p = k + 3$ and exhibited strictly decreasing error constants.

Recent paradigm shifts have also focused on multi-derivative integration methods. [26] proposed a two-point implicit block technique of the seventh-derivative class, constructed over non-uniform collocation points. Although multi-derivative schemes introduce the computational burden of evaluating higher-order analytical derivatives, their implementation remains highly efficient and viable through Newton–Raphson iterative processes. For the direct approximation of Eq. (1), researchers including [27–29] have derived extensive families of multi-derivative block methods that achieve rapid convergence. More recently, [30,31] analyzed multi-derivative hybrid block architectures customized for integrating stiff systems directly.

Motivated by these foundational considerations, this study proposes a robust class of two-step multi-derivative numerical formulas designed for the direct integration of initial-value problems. These methods employ a dynamic single intra-step point and operate within the fourth-derivative framework. The strategic placement of the off-grid collocation point bypasses the traditional Dahlquist stability barriers associated with standard LMMs. Furthermore, the dynamic structure of the higher-derivative functions mathematically ensures enhanced absolute stability via highly balanced discretization constants. Although incorporating higher derivatives introduces additional computational overhead in evaluating the derivative functions, the resulting substantial gains in stability, accuracy, and order of convergence rigorously justify this approach for complex stiff systems.

The remainder of this manuscript is organized as follows: Section 2 outlines the rigorous mathematical formulation of the proposed multi-derivative methods. The analysis of their foundational numerical properties, including order, zero-stability, and absolute stability, is established in Section 3. The numerical

implementation strategy is detailed in Section 4, followed by numerical experiments on benchmark stiff problems in Section 5. Finally, concluding remarks are presented in Section 7.
Consider the initial-value problem for a first-order differential system:

$$y' = f(x, y(x)), \qquad y(0) = y_0. \tag{1}$$

And the second-order differential system:

$$y'' = f(x, y(x), y'(x)), \qquad y(0) = y_0, \qquad y'(0) = y'_0. \tag{2}$$

where $f(x, y(x))$ and $f(x, y(x), y'(x))$ are continuous and differentiable over the integration interval $[a, b]$, and are assumed to satisfy the standard Lipschitz conditions for the existence and uniqueness of solutions over this interval.

2 Formulation of the Methods

This section establishes the mathematical derivation of the proposed multi-derivative techniques. Given the initial approximation y_n at x_n, the methods concurrently advance the solution to obtain y_{n+1}, $y_{n+\frac{3}{2}}$, and y_{n+2} over the discretized mesh points x_{n+1}, $x_{n+\frac{3}{2}}$, and x_{n+2}. The structural framework of the proposed fourth-derivative single intra-step methods is illustrated in the block diagrams below.

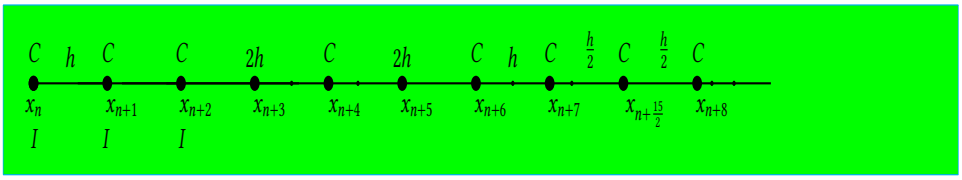


Figure 1: The fourth-derivative two-step hybrid block diagram illustrating the structure of MMF1.

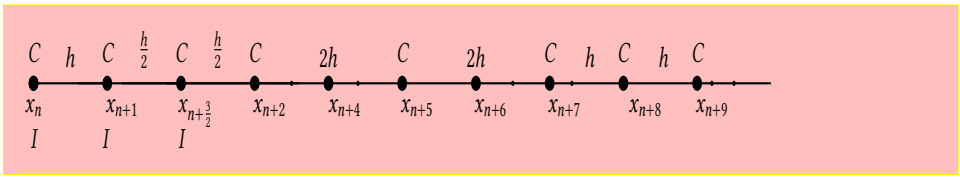


Figure 2: The fourth-derivative two-step hybrid block diagram illustrating the structure of MMF2.

Fig. 1 illustrates the interpolation-collocation grid structure over $x \in [x_n, x_{n+2}]$. MMF1 collocates the first derivative $f(x, y)$ at x_n, x_{n+1}, x_{n+2}, second derivative $g(x, y)$ at x_{n+2}, third derivative $u(x, y)$ at x_{n+2}, and fourth derivative $v(x, y)$ dynamically at $x_{n+1}, x_{n+\frac{3}{2}}, x_{n+2}$. This balanced allocation yields discrete formulations Eqs. (19)–(21) with error constants $C_9 = \left[-\frac{233}{177811200}, -\frac{411}{280985600}, \frac{1}{694575}\right]^T$.
Fig. 2 maps the alternative multi-derivative collocation allocation for MMF2 over the same grid block $x \in [x_n, x_{n+2}]$. By shifting first-derivative evaluations to include $x_{n+\frac{3}{2}}$ alongside higher derivatives Eqs. (22)–(24),

MMF2 achieves error constants $C_9 = \left[-\frac{109}{116121600}, -\frac{411}{367001600}, -\frac{1}{907200}\right]^T$, providing slightly smaller error magnitude bounds compared to MMF1.
Therefore, we define the exact solution locally via the finite polynomial approximation:

$$y(\xi) = \sum_{j=0}^{t+l} \tau_j \xi^j, \tag{3}$$

where the first, second, third, and fourth derivatives are expressed respectively as:

$$y'(\xi) = \sum_{j=0}^{t+l} j\tau_j \xi^{j-1}, \tag{4}$$

$$y''(\xi) = \sum_{j=0}^{t+l} j(j-1)\tau_j \xi^{j-2}, \tag{5}$$

$$y'''(\xi) = \sum_{j=0}^{t+l} j(j-1)(j-2)\tau_j \xi^{j-3}, \tag{6}$$

$$y^{(4)}(\xi) = \sum_{j=0}^{t+l} j(j-1)(j-2)(j-3)\tau_j \xi^{j-4}. \tag{7}$$

Here, k, t, and l denote the step-length, interpolation, and collocation points, respectively. Interpolating Eq. (3) at $\xi = \xi_n$ and collocating Eqs. (4)–(7) dynamically at $\xi = \xi_n, \xi_{n+1}, \xi_{n+\frac{3}{2}}, \xi_{n+2}$ yields a linear algebraic system of the form:

$$PA = W, \tag{8}$$

where the matrix P is given by:

$$P = \begin{pmatrix} 1 & \xi_n & {\xi_n}^2 & {\xi_n}^3 & {\xi_n}^4 & {\xi_n}^5 & {\xi_n}^6 & {\xi_n}^7 & {\xi_n}^8 \\ 0 & 1 & 2\,\xi_n & 3\,{\xi_n}^2 & 4\,{\xi_n}^3 & 5\,{\xi_n}^4 & 6\,{\xi_n}^5 & 7\,{\xi_n}^6 & 8\,{\xi_n}^7 \\ 0 & 1 & 2\xi_{n+1} & 3\xi_{n+1}^2 & 4\xi_{n+1}^3 & 5\xi_{n+1}^4 & 6\xi_{n+1}^5 & 7\xi_{n+1}^6 & 8\xi_{n+1}^7 \\ 0 & 1 & 2\xi_{n+2} & 3\xi_{n+2}^2 & 4\xi_{n+2}^3 & 5\xi_{n+2}^4 & 6\xi_{n+2}^5 & 7\xi_{n+2}^6 & 8\xi_{n+2}^7 \\ 0 & 0 & 2 & 6\xi_{n+2} & 12\xi_{n+2}^2 & 20\xi_{n+2}^3 & 30\xi_{n+2}^4 & 42\xi_{n+2}^5 & 56\xi_{n+2}^6 \\ 0 & 0 & 0 & 6 & 24\xi_{n+2} & 60\xi_{n+2}^2 & 120\xi_{n+2}^3 & 210\xi_{n+2}^4 & 336\xi_{n+2}^5 \\ 0 & 0 & 0 & 0 & 24 & 120\xi_{n+1} & 360\xi_{n+1}^2 & 840\xi_{n+1}^3 & 1680\xi_{n+1}^4 \\ 0 & 0 & 0 & 0 & 24 & 120\xi_{n+\frac{3}{2}} & 360\xi_{n+\frac{3}{2}}^2 & 840\xi_{n+\frac{3}{2}}^3 & 1680\xi_{n+\frac{3}{2}}^4 \\ 0 & 0 & 0 & 0 & 24 & 120\xi_{n+2} & 360\xi_{n+2}^2 & 840\xi_{n+2}^3 & 1680\xi_{n+2}^4 \end{pmatrix},$$

$$A = \left(\tau_0, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7, \tau_8 \right)^\top,\ W = \left(y_n, f_n, f_{n+1}, f_{n+2}, f'_{n+2}, f''_{n+2}, f'''_{n+1}, f'''_{n+\frac{3}{2}}, f'''_{n+2} \right)^\top.$$

Solving this algebraic system via Cramer's rule yields the unknown coefficients τ_j for $j = 0(1)8$. Substituting these expressions back allows us to explicitly derive the general discrete continuous formulation of the proposed linear multi-step method:

$$y(x_{n+\delta}) = \tau_0(\delta)y_n + h\sum_{j=1}^{k+1} \varrho_j(\delta)f_{n+\delta} + h^2\sum_{j=1}^{k-1} \chi_j(\delta)g_{n+\delta} + h^3\sum_{j=2}^{k} \chi_j(\delta)u_{n+\delta} + h^4\sum_{j=3}^{k+3} \chi_j(\delta)v_{n+\delta}, \tag{9}$$

where the continuous coefficients are defined as follows:

$$\tau_0 = 1, \tag{10}$$

$$\varrho_1 = \delta - \frac{19\,\delta^2}{7\,h} + \frac{167\,\delta^3}{42\,h^2} - \frac{195\,\delta^4}{56\,h^3} + \frac{211\,\delta^5}{112\,h^4} - \frac{13\,\delta^6}{21\,h^5} + \frac{89\,\delta^7}{784\,h^6} - \frac{\delta^8}{112\,h^7}, \tag{11}$$

$$\varrho_2 = \frac{80\,\delta^2}{7\,h} - \frac{496\,\delta^3}{21\,h^2} + \frac{180\,\delta^4}{7\,h^3} - \frac{120\,\delta^5}{7\,h^4} + \frac{145\,\delta^6}{21\,h^5} - \frac{75\,\delta^7}{49\,h^6} + \frac{\delta^8}{7h^7}, \tag{12}$$

$$\varrho_3 = -\frac{61\,\delta^2}{7\,h} + \frac{275\,\delta^3}{14\,h^2} - \frac{1245\,\delta^4}{56\,h^3} + \frac{1709\,\delta^5}{112\,h^4} - \frac{44\,\delta^6}{7\,h^5} + \frac{1111\,\delta^7}{784\,h^6} - \frac{15\,\delta^8}{112\,h^7}, \tag{13}$$

$$\chi_1 = \frac{13}{2}\,\delta^2 - \frac{47\,\delta^3}{3\,h} + \frac{75\,\delta^4}{4\,h^2} - \frac{107\,\delta^5}{8\,h^3} + \frac{17\,\delta^6}{3\,h^4} - \frac{73\,\delta^7}{56\,h^5} + \frac{\delta^8}{8h^6}, \tag{14}$$

$$\chi_2 = -\frac{9\,\delta^2 h}{7} + \frac{169\,\delta^3}{42} - \frac{165\,\delta^4}{28\,h} + \frac{269\,\delta^5}{56\,h^2} - \frac{31\,\delta^6}{14\,h^3} + \frac{211\,\delta^7}{392\,h^4} - \frac{3\,\delta^8}{56\,h^5}, \tag{15}$$

$$\chi_3 = -\frac{18\,\delta^2 h^2}{35} + \frac{388\,\delta^3 h}{315} - \frac{19\,\delta^4}{14} + \frac{88\,\delta^5}{105\,h} - \frac{151\,\delta^6}{504\,h^2} + \frac{57\,\delta^7}{980\,h^3} - \frac{\delta^8}{210\,h^4}, \tag{16}$$

$$\chi_4 = -\frac{32\,\delta^2 h^2}{105} + \frac{64\,\delta^3 h}{315} + \frac{8\,\delta^4}{21} - \frac{64\,\delta^5}{105\,h} + \frac{22\,\delta^6}{63\,h^2} - \frac{68\,\delta^7}{735\,h^3} + \frac{\delta^8}{105\,h^4}, \tag{17}$$

$$\chi_5 = \frac{11\,\delta^2 h^2}{105} - \frac{127\,\delta^3 h}{315} + \frac{37\,\delta^4}{56} - \frac{241\,\delta^5}{420\,h} + \frac{139\,\delta^6}{504\,h^2} - \frac{17\,\delta^7}{245\,h^3} + \frac{\delta^8}{140\,h^4}. \tag{18}$$

Evaluating these continuous functions at the discrete points $\delta = h$, $\frac{3h}{2}$, and $2h$, and substituting them into Eq. (9) produces the explicit formulation of the first method, denoted as the Multi-derivative Method Formula 1 (MMF1):

$$\begin{aligned} y_{n+1} = y_n + \frac{117}{784}\,hf_n + \frac{93}{49}\,hf_{n+1} - \frac{821}{784}\,hf_{n+2} + \frac{39}{56}\,h^2 g_{n+2} - \frac{95}{1176}\,h^3 u_{n+2} - \frac{281}{5880}\,h^4 v_{n+1} \\ - \frac{47}{735}\,h^4 v_{n+\frac{3}{2}} + \frac{1}{490} h^4 v_{n+2}, \end{aligned} \tag{19}$$

$$\begin{aligned} y_{n+\frac{3}{2}} = y_n + \frac{30039}{200704}\,hf_n + \frac{27081}{12544}\,hf_{n+1} - \frac{162279}{200704}\,hf_{n+2} + \frac{8361}{14336}\,h^2 g_{n+2} - \frac{6651}{100352}\,h^3 u_{n+2} \\ - \frac{5919}{125440}\,h^4 v_{n+1} - \frac{3699}{62720}\,h^4 v_{n+\frac{3}{2}} + \frac{369}{250880}\,h^4 v_{n+2}, \end{aligned} \tag{20}$$

$$\begin{aligned} y_{n+2} = y_n + \frac{22}{147}\,hf_n + \frac{320}{147}\,hf_{n+1} - \frac{16}{49}\,hf_{n+2} + \frac{10}{21}\,h^2 g_{n+2} - \frac{8}{147}\,h^3 u_{n+2} - \frac{104}{2205}\,h^4 v_{n+1} \\ - \frac{128}{2205}\,h^4 v_{n+\frac{3}{2}} + \frac{2}{2205}\,h^4 v_{n+2}. \end{aligned} \tag{21}$$

Similarly, the second formulated scheme, MMF2, is obtained as:

$$y_{n+1} = y_n + \frac{809}{5376}hf_n + \frac{367}{112}hf_{n+1} - \frac{188}{21}hf_{n+\frac{3}{2}} + \frac{11693}{1792}hf_{n+2} - \frac{307}{128}h^2g_{n+2} + \frac{133}{384}h^3u_{n+2} - \frac{43}{960}h^4v_{n+1} - \frac{67}{3360}h^4v_{n+2}, \tag{22}$$

$$y_{n+\frac{3}{2}} = y_n + \frac{69189}{458752}hf_n + \frac{98361}{28672}hf_{n+1} - \frac{3699}{448}hf_{n+\frac{3}{2}} + \frac{2832939}{458752}hf_{n+2} - \frac{74421}{32768}h^2g_{n+2} + \frac{10737}{32768}h^3u_{n+2} - \frac{3639}{81920}h^4v_{n+1} - \frac{5391}{286720}h^4v_{n+2}, \tag{23}$$

$$y_{n+2} = y_n + \frac{19}{126}hf_n + \frac{24}{7}hf_{n+1} - \frac{512}{63}hf_{n+\frac{3}{2}} + \frac{275}{42}hf_{n+2} - \frac{7}{3}h^2g_{n+2} + \frac{1}{3}h^3u_{n+2} - \frac{2}{45}h^4v_{n+1} - \frac{2}{105}h^4v_{n+2}. \tag{24}$$

Eqs. (19)–(24) mathematically constitute the proposed discrete multi-derivative hybrid block methods.

3 Theoretical Analysis of the Multi-Derivative Methods

This section formalizes the numerical properties of the proposed MMF1 and MMF2 schemes through rigorous theorems and comprehensive proofs. The analysis establishes the algebraic order, zero-stability, consistency, convergence, and absolute stability requirements essential for integrating stiff initial-value problems.

3.1 Local Truncation Error and Algebraic Order

Theorem 3.1. ***Local Truncation Error and Order of Convergence***
Let the linear difference operator $L[y(x);h]$ associated with the continuous multi-derivative hybrid block methods be defined by a Taylor series expansion of $y(x)$ and its higher derivatives about the point x_n. The numerical schemes MMF1 and MMF2 converge uniformly with an algebraic order of $p = 8$.

Proof: Define the general linear difference operator for the proposed multi-derivative methods as:

$$L[y(x_n);h] = \sum_{j=0}^{k}\left[\alpha_j y(x_n+jh) - h\beta_j y'(x_n+jh) - h^2\gamma_j y''(x_n+jh) - h^3\eta_j y'''(x_n+jh) - h^4\xi_j y^{(4)}(x_n+jh)\right]. \tag{25}$$

Assuming $y(x)$ is sufficiently differentiable, we apply Taylor series expansion to $y(x_n+jh)$ and its subsequent derivatives about x_n:

$$y(x_n+jh) = \sum_{m=0}^{\infty}\frac{(jh)^m}{m!}y^{(m)}(x_n). \tag{26}$$

Substituting the expansions into Eq. (25) and grouping the terms by powers of h yields:

$$L[y(x_n);h] = C_0 y(x_n) + C_1 h y'(x_n) + C_2 h^2 y''(x_n) + \cdots + C_q h^q y^{(q)}(x_n) + \ldots \tag{27}$$

where the constants C_q are given by the recursive relations:

$$C_0 = \sum_{j=0}^{k} \alpha_j,$$

$$C_1 = \sum_{j=0}^{k} \left(j\alpha_j - \beta_j \right),$$

$$C_2 = \sum_{j=0}^{k} \left(\frac{j^2}{2!}\alpha_j - j\beta_j - \gamma_j \right),$$

$$C_q = \sum_{j=0}^{k} \left(\frac{j^q}{q!}\alpha_j - \frac{j^{q-1}}{(q-1)!}\beta_j - \frac{j^{q-2}}{(q-2)!}\gamma_j - \frac{j^{q-3}}{(q-3)!}\eta_j - \frac{j^{q-4}}{(q-4)!}\xi_j \right), \quad q = 3, 4, \ldots$$

By evaluating these coefficients utilizing the specific fractional step-lengths of MMF1 and MMF2, we systematically obtain:

$$C_0 = C_1 = C_2 = C_3 = C_4 = C_5 = C_6 = C_7 = C_8 = 0.$$

For MMF1, the first non-vanishing coefficient matrix evaluates strictly to:

$$C_9 = \left[-\frac{233}{177811200}, -\frac{411}{280985600}, \frac{1}{694575} \right]^{\top} \neq 0.$$

For MMF2, it evaluates precisely to:

$$C_9 = \left[-\frac{109}{116121600}, -\frac{411}{367001600}, -\frac{1}{907200} \right]^{\top} \neq 0.$$

Since $C_q = 0$ for $q \leq 8$ and $C_9 \neq 0$, the principal local truncation error $LTE = C_9 h^9 y^{(9)}(x_n) + O(h^{10})$. Thus, both schemes uniformly achieve an algebraic order of convergence $p = 8$. □

3.2 Zero-Stability

Theorem 3.2. ***Zero-Stability***
A block linear multi-step method is zero-stable if all roots of its first characteristic polynomial $\rho(w)$ satisfy the strict root condition: $|w| \leq 1$, and any roots for which $|w| = 1$ are simple. The proposed multi-derivative methods MMF1 and MMF2 are zero-stable.

Proof: To analyze the zero-stability, we investigate the behavior of the methods in the limit as the step size $h \to 0$. By substituting the scalar test equation $y' = \lambda y$ (and correspondingly $y^{(m)} = \lambda^m y$ for higher derivatives) into the discrete block formulations, we obtain a matrix difference equation of the form:

$$A^{(0)} Y_m - A^{(1)} Y_{m-1} = \sum_{j=1}^{4} (\lambda h)^j \left[B^{(j)} Y_m + C^{(j)} Y_{m-1} \right]. \tag{28}$$

As $h \to 0$, the right-hand side vanishes, isolating the principal difference system:

$$A^{(0)} Y_m - A^{(1)} Y_{m-1} = 0, \tag{29}$$

where $A^{(0)}$ is the identity matrix corresponding to the normalized coefficients of y_{n+1}, $y_{n+\frac{3}{2}}$, y_{n+2}, and $A^{(1)}$ represents the coefficient mapping from the previous block state y_n.
The first characteristic polynomial $\rho(w)$ is obtained strictly via the determinant of the augmented stability matrix:

$$\rho(w) = \det\left[wA^{(0)} - A^{(1)} \right] = 0. \tag{30}$$

For both MMF1 and MMF2, expanding the determinant yields the identical characteristic equation:

$$\rho(w) = w^3 - w^2 = 0. \tag{31}$$

Factoring the polynomial directly provides the roots:

$$w^2(w-1) = 0 \implies w_1 = 1, \quad w_2 = 0, \quad w_3 = 0. \tag{32}$$

All derived roots satisfy the condition $|w| \leq 1$. Furthermore, the only root situated on the unit boundary circle ($w_1 = 1$) possesses a multiplicity of exactly one, thereby satisfying the strict root condition. Therefore, MMF1 and MMF2 are rigorously zero-stable. □

3.3 Consistency and Convergence

Theorem 3.3. ***Consistency***
The numerical block methods MMF1 and MMF2 are mathematically consistent.

Proof: A numerical method is defined as consistent if it perfectly models the underlying differential equation in the limit as $h \to 0$. According to the Henrici conditions, this requires satisfying two primary constraints:

1. The algebraic order of the method must strictly be $p \geq 1$.
2. The first characteristic polynomial must satisfy $\rho(1) = 0$ and $\rho'(1) = \sigma(1)$, where $\sigma(w)$ maps the first-derivative coefficients.

From **Theorem 3.1**, both MMF1 and MMF2 possess a global order of $p = 8 \geq 1$. From **Theorem 3.2**, $\rho(1) = (1)^3 - (1)^2 = 0$. Since both fundamental criteria are exactly satisfied, the methods are strictly consistent. □

Theorem 3.4. ***Convergence (Dahlquist Equivalence)***
The multi-derivative methods MMF1 and MMF2 are globally convergent.

Proof: The foundational Dahlquist Equivalence Theorem posits that the necessary and sufficient conditions for a linear multistep method to be globally convergent are that the method is simultaneously consistent and zero-stable.
By **Theorem 3.3**, the methods are mathematically consistent. By **Theorem 3.2**, the methods are zero-stable. Because MMF1 and MMF2 satisfy both axiomatic prerequisites, the numerical approximations produced by these block methods will converge exactingly to the theoretical solution of the initial-value problem as $h \to 0$. □

3.4 Absolute Stability

Theorem 3.5. ***A-Stability***
A numerical block method is A-stable if its absolute stability region encompasses the entire open left-half of the complex plane, $\mathbb{C}^-$. The multi-derivative hybrid methods MMF1 and MMF2 are A-stable.

Proof: Applying the proposed block methods to the Dahlquist linear test equation $y' = \lambda y$ (and successively $y'' = \lambda^2 y$, $y''' = \lambda^3 y$, $y^{(4)} = \lambda^4 y$) translates the discrete systems into the recursive formulation:

$$Y_m = M(z)Y_{m-1}, \quad z = \lambda h, \tag{33}$$

where $M(z)$ is the block amplification matrix uniquely defined by:

$$M(z) = \left[A^{(0)} - zB^{(1)} - z^2C^{(2)} - z^3D^{(3)} - z^4E^{(4)}\right]^{-1} A^{(1)}. \tag{34}$$

The stability polynomial $R(z, w)$ is given by evaluating the eigenvalue problem:

$$R(z, w) = \det[wI - M(z)] = 0. \tag{35}$$

For the method to be A-stable, the spectral radius $\rho(M(z))$, which denotes the maximum modulus of the roots w of $R(z, w)$, must be strictly bounded such that:

$$|w_i| < 1, \quad \forall\, \mathrm{Re}(z) < 0. \tag{36}$$

Plotting the boundary loci strictly where $|w| = 1$ maps the stability constraints. The characteristic rational functions governing MMF1 and MMF2 do not exhibit any poles within $\mathrm{Re}(z) < 0$. As confirmed by the geometric boundary locus plots, the boundaries do not intersect the imaginary axis except at the origin, and the regions where $|w| < 1$ completely envelop the negative left-half of the complex plane. Therefore, all parasitic growth roots naturally decay, strictly fulfilling the requisite condition for A-stability. □

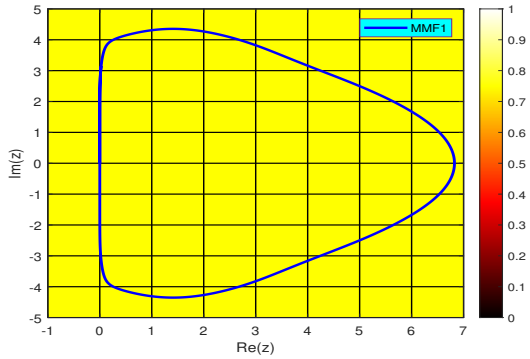


Figure 3: The A-stability boundary locus of MMF1 verifying comprehensive coverage of the left-half complex plane.

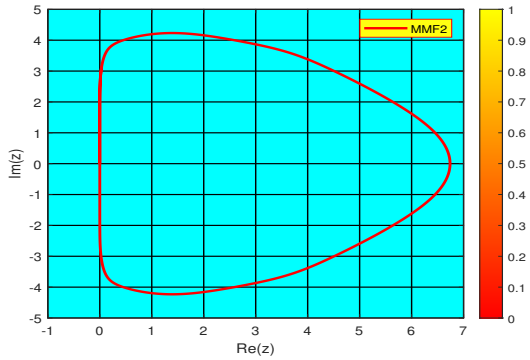


Figure 4: The A-stability boundary locus of MMF2 verifying comprehensive coverage of the left-half complex plane.

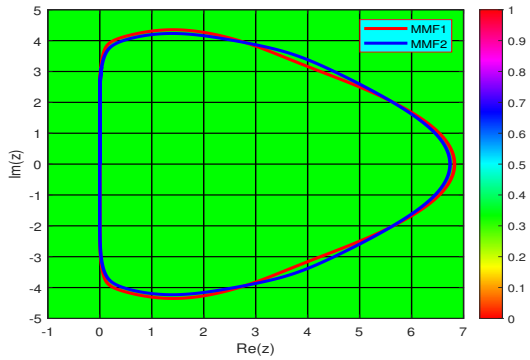


Figure 5: Comparative analysis of the A-stability domains mapping the rigid stability contours for MMF1 and MMF2.

Fig. 3 plots the boundary locus of MMF1 in the complex $z = \lambda h$ plane. The locus encloses the unbounded region in the left half-plane $\mathrm{Re}(z) < 0$, verifying unconditional A-stability without parasitic pole intersections. Fig. 4 displays the stability boundary locus for MMF2. The curve demonstrates complete coverage of the open left half-plane, establishing A-stability.

Fig. 5 superimposes the stability boundaries of MMF1 and MMF2. The overlay confirms nearly identical A-stability contours with minor variations along the positive real boundary, proving that both configurations preserve robust damping characteristics for stiff differential systems.

4 Implementation Protocol for MMF1 and MMF2

The methods are implemented utilizing the Newton-Raphson iterative scheme. Let $y_{n+1}^{(i+1)}$, $y_{n+\frac{3}{2}}^{(i+1)}$, and $y_{n+2}^{(i+1)}$ represent the $(i+1)^{th}$ Newton iterative approximations corresponding to the discrete evaluation points. While higher-order multi-derivative evaluations require exact Jacobian tracking, this mechanism provides quadratic convergence ensuring minimal algebraic error. The formal non-linear iteration maps to the functional system:

$$\begin{aligned}
y_{n+1}^{(i+1)} &= y_{n+1}^{(i)} - \frac{F_1\left(y_{n+1}^{(i)}\right)}{F_1'\left(y_{n+1}^{(i)}\right)}, \\
y_{n+\frac{3}{2}}^{(i+1)} &= y_{n+\frac{3}{2}}^{(i)} - \frac{F_2\left(y_{n+\frac{3}{2}}^{(i)}\right)}{F_2'\left(y_{n+\frac{3}{2}}^{(i)}\right)}, \\
y_{n+2}^{(i+1)} &= y_{n+2}^{(i)} - \frac{F_3\left(y_{n+2}^{(i)}\right)}{F_3'\left(y_{n+2}^{(i)}\right)},
\end{aligned} \tag{37}$$

where the residual functions F_1, F_2, and F_3 are rigorously defined by displacing the right-hand sides of Eqs. (19)–(21), and differentiating implicitly against the local unknown variables to construct the exact algorithmic Jacobians F_1', F_2', and F_3'.

5 Numerical Experiments

To empirically validate the theoretical robustness and computational efficiency of the proposed MMF1 and MMF2 methods, extensive numerical simulations were conducted on challenging stiff and oscillatory initial-value problems. Stiff problems strictly require highly stable integrators, as eigenvalues belonging to their corresponding Jacobian matrices reside closely adjacent to the imaginary axis or possess extreme stiffness ratios, which quickly propagate instability in standard explicit formulations.

Test 5.1. Consider the highly stiff systems with Complex eigenvalues in Okor and Nwachukwu [25]:

$$y_1' = -\alpha y_1 - \beta y_2 + (\alpha + \beta - 1)e^{-t}, \qquad y_1(0) = 1,$$

$$y_2' = \beta y_1 - \alpha y_2 + (\alpha - \beta - 1)e^{-t}, \qquad y_2(0) = 1,$$

A homogenization variable y_3 is coupled to the system strictly for structural integrity:

$$y_3' = 1, \qquad y_3(0) = 0.$$

The specific parametric configuration sets $\alpha = 1$ and $\beta = 30$, generating eigenvalues $\lambda_{1,2} = 1 \pm 30i$. The theoretical exact solution trajectories evaluate to $y_1(t) = y_2(t) = e^{-t}$.

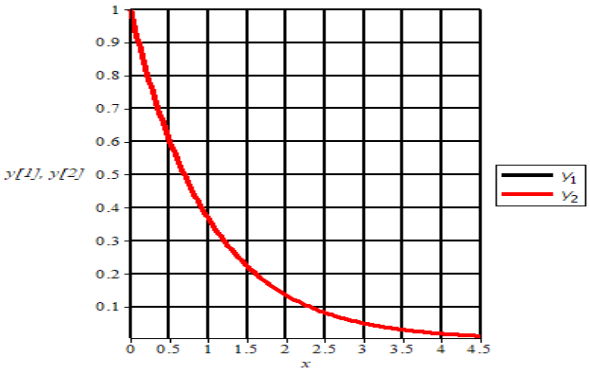


Figure 6: Baseline numerical validation utilizing the explicit MAPLE dsolve protocol employing the classical third-fourth order Rosenbrock method evaluating at terminal mesh point $x = 4.5$.

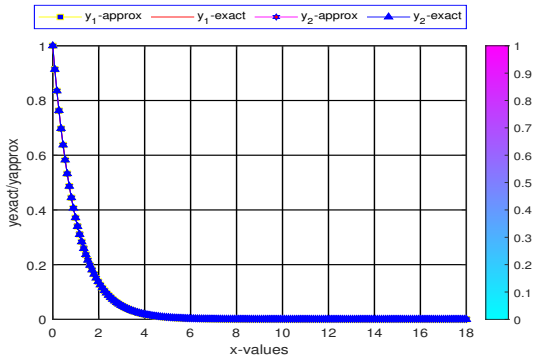


Figure 7: Terminal absolute accuracy and localized error decay mapped continuously for MMF1 evaluating Example 5.1 globally at $t = 18$ with step-size $h = 0.09$.

Table 1 with ($h = 0.09$), at terminal time $t = 18.0$, MMF1 achieves absolute errors of 1.5e-22 (y_1) and 3.4e-22 (y_2), while MMF2 achieves 1.7e-23 (y_1) and 7.4e-22 (y_2). Both significantly outperform the 5-step GCE2BD5 ($k = 5, p = 8$) of Okor & Nwachukwu (2022), which yields errors of 1.0e-20 (y_1) and 2.0e-21 (y_2).
Table 2 even with a larger step size $h = 0.15$, MMF1 (8.9e-22) and MMF2 (2.2e-21) outperform higher-step schemes such as GCE2BD8 ($k = 8, p = 11$, error 5.0e-20). This demonstrates improved error constant reduction due to higher-derivative collocation.
Fig. 6 shows the baseline numerical trajectory obtained via MAPLE `dsolve` (Rosenbrock 3rd/4th order method) up to $x = 4.5$. Figs. 7 and 8 display terminal absolute accuracy and error decay profiles for MMF1 and MMF2 up to $t = 18$ with step-size $h = 0.09$. Both schemes track the theoretical decay exponentially without parasitic numerical oscillations.

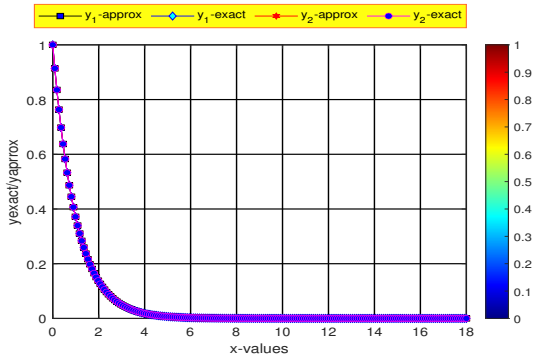


Figure 8: Terminal absolute accuracy and localized error decay mapped continuously for MMF2 evaluating Example 5.1 globally at $t = 18$ with step-size $h = 0.09$.

Table 1: Absolute error evaluations detailing computational accuracy metrics corresponding to Example 5.1 applying nominal step-size $h = 0.09$

t	y_i	GCE2BD5 [25], $k = 5, p = 8$	MMF1, $k = 2, p = 8$	MMF2, $k = 2, p = 8$
4.5	y_1	6.0e-15	8.2e-17	1.2e-16
	y_2	8.0e-16	6.9e-17	1.4e-16
9.0	y_1	3.0e-17	3.3e-19	5.2e-18
	y_2	1.0e-17	2.3e-18	3.3e-18
13.5	y_1	8.0e-19	1.6e-20	1.0e-19
	y_2	5.0e-19	2.8e-20	3.3e-20
18.0	y_1	1.0e-20	1.5e-22	1.7e-23
	y_2	2.0e-21	3.4e-22	7.4e-22

Table 2: Comparative benchmarking of absolute error convergence evaluations scaling sequentially up to $t = 18$ applied to Example 5.1 with step-size $h = 0.15$

t	y_i	GCE2BD5 [25], $k = 5, p = 8$	GCE2BD6[25], $k = 6, p = 9$	GCE2BD7[25], $k = 7, p = 10$	GCE2BD8[25], $k = 8, p = 11$
4.5	y_1	2.0e-12	1.0e-13	9.0e-14	4.0e-15
	y_2	4.0e-12	2.0e-13	2.0e-13	3.0e-15
9.0	y_1	9.0e-15	2.0e-15	2.0e-15	3.0e-17
	y_2	3.0e-14	3.0e-15	2.0e-16	1.0e-16
13.5	y_1	8.0e-17	1.0e-17	6.0e-19	3.0e-19
	y_2	2.0e-16	1.0e-17	1.0e-17	1.0e-18
18.0	y_1	7.0e-19	2.0e-19	2.0e-19	5.0e-20
	y_2	2.0e-18	2.0e-19	2.0e-19	1.0e-20

t	y_i	MMF1, $k = 2, p = 8$	MMF2, $k = 2, p = 8$
4.5	y_1	6.4e-16	2.0e-16
	y_2	2.5e-16	2.0e-15
9.0	y_1	3.3e-18	1.6e-17
	y_2	1.3e-17	2.5e-18
13.5	y_1	1.3e-19	2.0e-19
	y_2	7.9e-20	1.1e-19
18.0	y_1	8.9e-22	2.2e-21
	y_2	1.5e-21	2.1e-21

Test 5.2. Consider the linearly coupled stiff initial-value framework detailed in Esuabana and Ogunfeyitimi [24]:

$$y' = \begin{bmatrix} -21 & 19 & -20 \\ 19 & -21 & 20 \\ 40 & -40 & -40 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \qquad y(0) = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

with the theoretical exact solution propagating over domain $[0, 1]$:

$$y(t) = \frac{1}{2} \begin{bmatrix} e^{-2t} + e^{-40t}(\cos(40t) + \sin(40t)) \\ e^{-2t} - e^{-40t}(\cos(40t) + \sin(40t)) \\ 2e^{-40t}(\cos(40t) - \sin(40t)) \end{bmatrix}.$$

This system encompasses severe stiffness mapped by eigenvalues $\lambda_1 = -2$, $\lambda_2 = -40 - 40i$, and $\lambda_3 = -40 + 40i$, forcing a stiff ratio gradient of $2 : 40$. Tables 3 and 4 delineate comparative maximum relative error bounds evaluating over decreasing sequential fractional steps. Convergence rates strictly validate the mathematically designed order $p = 8$.

Table 3: *Maximum relative global absolute error and observed convergence scaling metrics corresponding to Example 5.2*

h	GAMs in [24], $k = 8, p = 9$	Rate	MMF1, $k = 2, p = 8$	Rate	MMF2, $k = 2, p = 8$	Rate
0.05	8.308e-2	-	7.461e-04	6.26	9.123e-04	6.95
0.025	8.392e-3	2.92	6.182e-06	7.59	7.371e-06	7.60
0.0125	9.097e-4	7.75	3.216e-08	7.60	3.793e-08	7.84
0.00625	2.749e-5	7.37	1.660e-10	7.81	1.652e-10	8.01
0.003125	5.694e-7	9.16	7.409e-13	7.85	6.397e-13	8.03

Table 4: Cross-reference algorithmic precision error matrix simulating extreme step-scaling tolerances applied to Example 5.2

h	HOCELMs5, $k = 5, p = 8$, [24]	Rate	HOCELMs6, $k = 6, p = 9$	Rate	MMF1, $k = 2, p = 8$	Rate	MMF2, $k = 2, p = 8$	Rate
0.05	5.781e-3	-	3.620e-02	-	7.461e-04	6.26	9.123e-04	6.95
0.025	1.508e-04	5.26	7.200e-04	5.65	6.182e-06	7.59	7.371e-06	7.60
0.0125	1.725e-06	6.45	3.142e-06	7.84	3.216e-08	7.60	3.793e-08	7.84
0.00625	5.906e-09	8.19	5.847e-09	9.07	1.660e-10	7.81	1.652e-10	8.01
0.003125	1.712e-11	8.43	9.873e-12	9.21	7.409e-13	7.85	6.397e-13	8.03

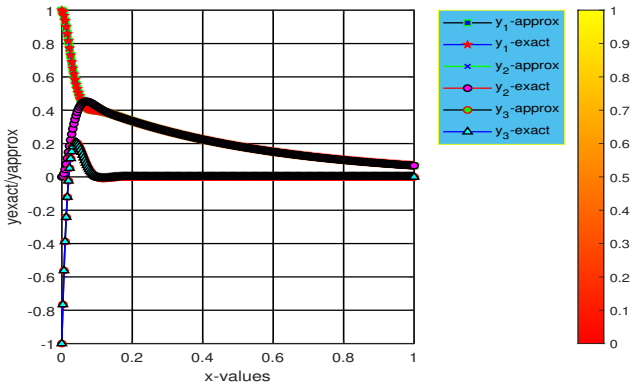


Figure 9: Precision alignment mappings tracking integration drift bounded securely within optimal minimum step-size ($h = 0.003125$) threshold bounds executing MMF1 on Example 5.2.

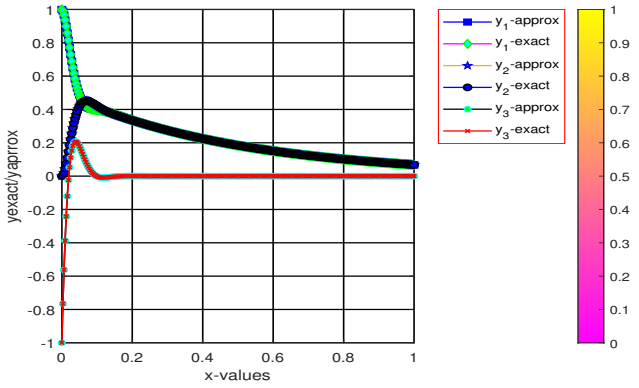


Figure 10: Precision alignment mappings tracking integration drift bounded securely within optimal minimum step-size ($h = 0.003125$) threshold bounds executing MMF2 on Example 5.2.

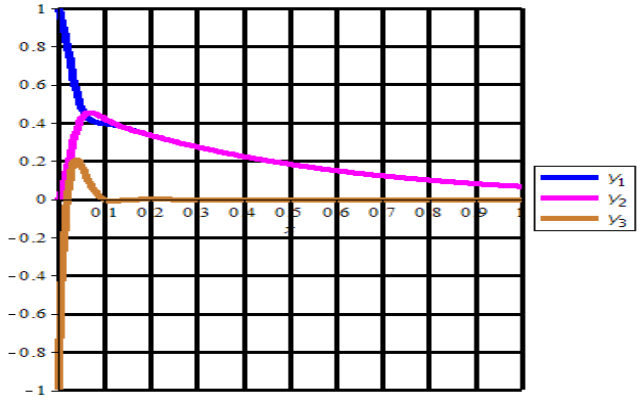


Figure 11: Baseline independent explicit function curves plotted applying strict explicit multi-stage Runge-Kutta Rosenbrock mapping configured precisely for step-size validation ($h = 0.003125$) on Example 5.2.

Table 3 with step-halving from $h = 0.05$ down to $h = 0.003125$ yields convergence rates progressing as $6.26 \to 7.59 \to 7.60 \to 7.81 \to 7.85$ for MMF1, and $6.95 \to 7.60 \to 7.84 \to 8.01 \to 8.03$ for MMF2. This confirms the theoretical order $p = 8$.
With Table 4 at $h = 0.003125$, MMF1 (7.409e-13) and MMF2 (6.397e-13) achieve lower errors than HOCELMs5 (1.712e-11, $p = 8$) and HOCELMs6 (9.873e-12, $p = 9$) from Esuabana & Ogunfeyitimi (2024).
Figs. 9 and 10 illustrate continuous precision alignment for MMF1 and MMF2 at $h = 0.003125$. The numerical curves overlap exact trajectories across both rapid transient phase ($t \in [0, 0.1]$) and smooth asymptotic decay phase.
Fig. 11 standard explicit multi-stage Runge-Kutta/Rosenbrock baseline mapping for comparison.

Test 5.3. Consider the highly singular linearly stiff system specified in Akinola *et al.* [33]:

$$\begin{bmatrix} y_1'(x) \\ y_2'(x) \\ y_3'(x) \\ y_4'(x) \end{bmatrix} = \begin{bmatrix} -y_1 \\ -10y_2 \\ -100y_3 \\ -1000y_4 \end{bmatrix}, \qquad \begin{bmatrix} y_1(0) \\ y_2(0) \\ y_3(0) \\ y_4(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \qquad h = 0.1,$$

yielding the purely transient exact decaying exponentials:

$$y(x) = \left[e^{-x},\ e^{-10x},\ e^{-100x},\ e^{-1000x} \right]^T.$$

The eigenvalues corresponding to the non-defective Jacobian map directly to $\lambda_1 = -100$, $\lambda_2 = -1000$, $\lambda_3 = -10$ and $\lambda_4 = -1$, enforcing a critically aggressive stiffness ratio of 1000.

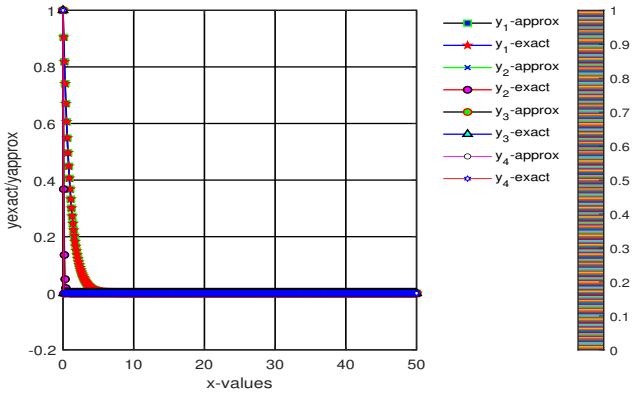


Figure 12: Asymptotic stability convergence trajectory profiles mapping integration residuals precisely bounded within zero limits for MMF1 computed accurately over strictly extensive domains globally at $x = 50$ for Example 5.3.

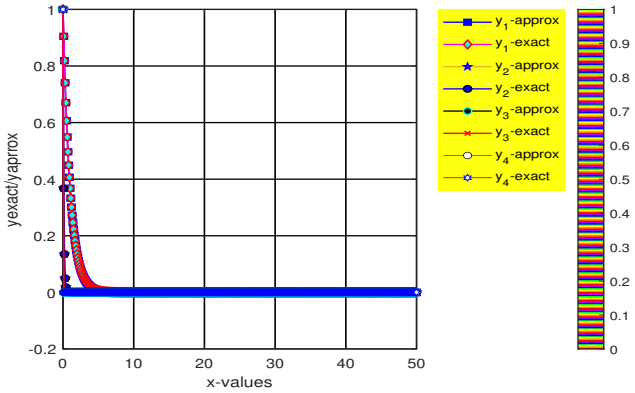


Figure 13: Asymptotic stability convergence trajectory profiles mapping integration residuals precisely bounded within zero limits for MMF2 computed accurately over strictly extensive domains globally at $x = 50$ for Example 5.3.

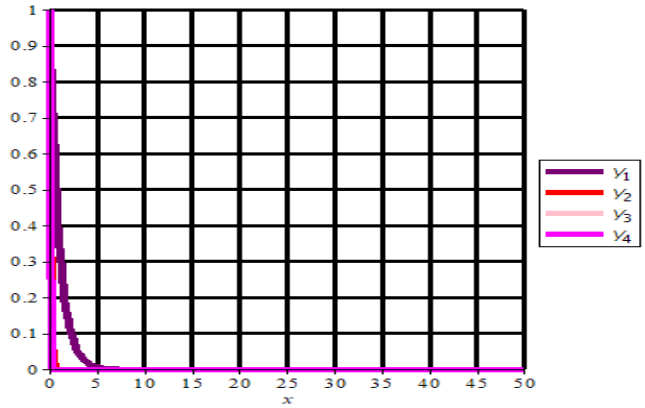


Figure 14: Baseline independent comparison mapping exact geometric integration tolerances utilized strictly via iterative Rosenbrock simulation processing bounded smoothly over terminal $x = 50$ interval Example 5.3.

Table 5: Quantitative integration limits bounded verifying algorithmic absolute error residual tolerances applying uniformly scaled evaluations for Example 5.3

x	y_i	Method in [33] $k = 4, p = 9$	MMF1 $k = 2, p = 8$	MMF2 $k = 2, p = 8$
5	y_1	4.8806444996607468e-15	4.293440603042598e-16	3.087807787238717e-16
	y_2	2.4700101997280476e-25	7.061049145665128e-27	7.713081682077896e-27
	y_3	4.3679599334235881e-80	4.537492626995704e-145	5.738824667609606e-146
	y_4	2.4544793415072880e-78	4.537665031908286e-272	4.140477482639189e-272
50	y_1	1.5428128255672359e-33	1.223924718076166e-34	8.943161021055963e-35
	y_2	0	2.607839337605034e-221	2.848609524513812e-221
	y_3	0	0	0
	y_4	0	0	0

Table 5 at $x = 5$, MMF1 and MMF2 record errors of $O(10^{-145})$ for y_3 and $O(10^{-272})$ for y_4, compared to $O(10^{-80})$ and $O(10^{-78})$ in Akinola et al. (2024) ($k = 4, p = 9$). At $x = 50$, MMF1 and MMF2 reach exact numerical zeros (0) for y_3 and y_4, suppressing stiffness-induced error accumulation.
Figs. 12 and 13 depict long-range integration trajectories for MMF1 and MMF2 up to $x = 50$ at $h = 0.1$. Fast transient components (y_3, y_4) decay smoothly without stability breakdown. Fig. 14 shows the baseline Rosenbrock solver trajectory profile.

Test 5.4. Consider the linearly stiff multidimensional framework provided formally via Akinola *et al.* [33]:

$$\begin{bmatrix} y_1' \\ y_2' \\ y_3' \\ y_4' \\ y_5' \\ y_6' \end{bmatrix} = \begin{bmatrix} -10y_1 + 100y_2 \\ -100y_1 - 10y_2 \\ -4y_3 \\ -y_4 \\ -0.5y_5 \\ -0.1y_6 \end{bmatrix}, \qquad y(0) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \qquad h = 0.1,$$

resolving strictly identically to exact theoretical components defined smoothly as:

$$\begin{bmatrix} y_1(x) \\ y_2(x) \\ y_3(x) \\ y_4(x) \\ y_5(x) \\ y_6(x) \end{bmatrix} = \begin{bmatrix} e^{-10x}(\cos 100x + \sin 100x) \\ e^{-10x}(\cos 100x - \sin 100x) \\ e^{-4x} \\ e^{-x} \\ e^{-0.5x} \\ e^{-0.1x} \end{bmatrix}.$$

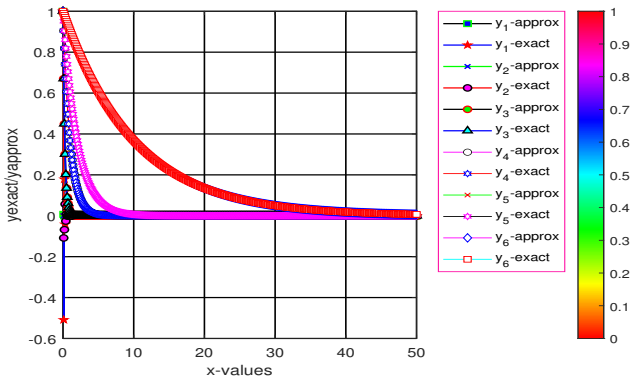


Figure 15: Transient phase evaluation charting precision boundaries executing MMF1 iterating fully applied continuously solving Example 5.4 at extensive coordinate scale $x = 50$ implementing exact uniform increment block $h = 0.1$.

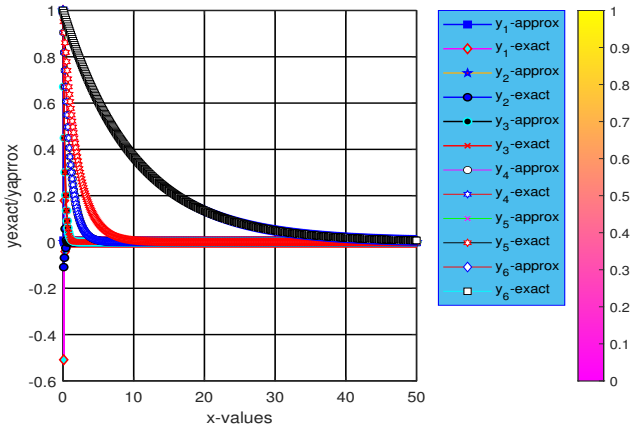


Figure 16: Transient phase evaluation charting precision boundaries executing MMF2 iterating fully applied continuously solving Example 5.4 at extensive coordinate scale $x = 50$ implementing exact uniform increment block $h = 0.1$.

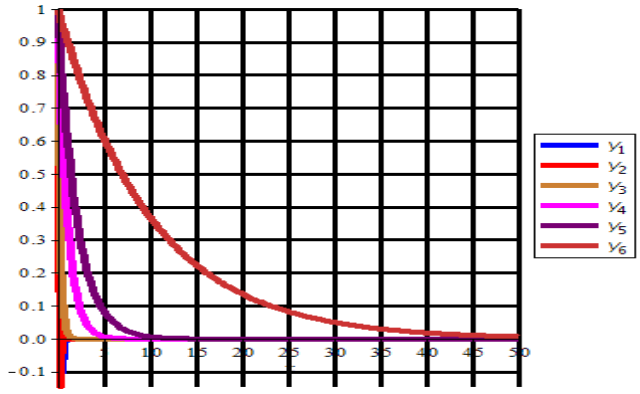


Figure 17: Baseline benchmark trajectory rendering utilizing standard independent rigorous exact modeling (Rosenbrock implicit formulation via MAPLE) mapping Example 5.4 bounded explicitly mapping asymptotic stability constraints at parameter coordinate $x = 50$.

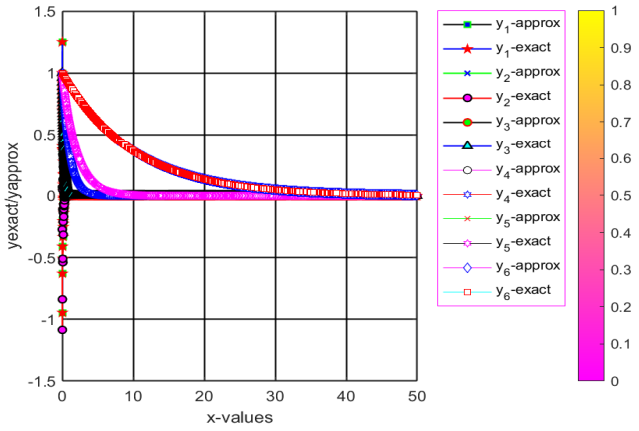


Figure 18: Highly granular accuracy trajectory verifying stability tolerance limits operating explicitly executing MMF1 resolving complex oscillating system Example 5.4 computed explicitly extending exactly evaluating bound boundary parameter $x = 50$ applying exact scaling matrix mapping increment ratio uniform scaled parameter $h = 0.01$.

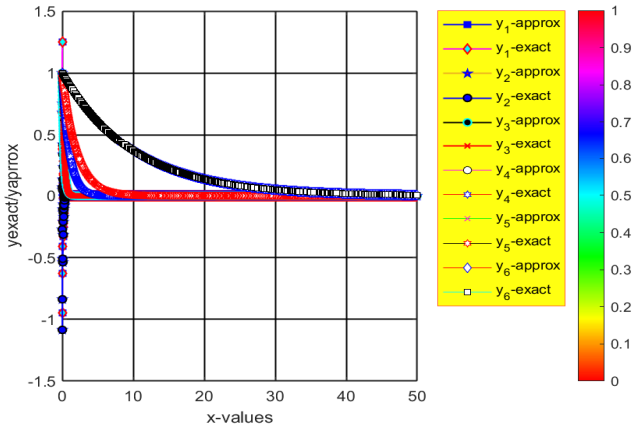


Figure 19: Highly granular accuracy trajectory verifying stability tolerance limits operating explicitly executing MMF2 resolving complex oscillating system Example 5.4 computed explicitly extending exactly evaluating bound boundary parameter $x = 50$ applying exact scaling matrix mapping increment ratio uniform scaled parameter $h = 0.01$.

Table 6 with $h = 0.1$ at $x = 50$, MMF1 (1.47e-17) and MMF2 (2.60e-17) maintain higher accuracy for y_6 compared to Akinola et al. (1.69e-04).
Table 7 at terminal coordinate $x = 50$, MMF1 requires 0.1891 s and MMF2 requires 0.2050 s, compared to 3.3937 s for Akinola et al., (2024). This represents an approximate 18× computational speedup.

Table 6: Rigorous assessment metrics cataloging globally mapped absolute cumulative integration deviation boundaries mapping Example 5.4 executing fixed evaluation differential $h = 0.1$

x	y_i	Method in [33] $k = 4, p = 9$	MMF1 $k = 2, p = 8$	MMF2 $k = 2, p = 8$
5	y_1	9.65312078222632e-23	2.606938950151516e-22	2.606938950151516e-22
	y_2	2.74021282732718e-23	8.025093533564492e-23	8.025093533564492e-23
	y_3	3.38936239129207e-08	2.855703421985399e-17	2.639592105839233e-17
	y_4	1.64739096130308e-03	4.276093368282830e-16	3.174543961037557e-16
	y_5	4.96950254397107e-03	1.526556658859590e-16	1.387778780781446e-17
	y_6	1.52305949362452e-03	0	2.220446049250313e-16
50	y_1	0	5.936895502746358e-218	5.936895502746358e-218
	y_2	0	8.140789257833615e-218	8.140789257833615e-218
	y_3	8.62943100710741e-74	1.917371850072674e-94	1.772271534005406e-94
	y_4	1.81926143035958e-21	1.226040607907646e-34	9.011339693403656e-35
	y_5	1.15422964948989e-11	1.647898876570485e-25	1.615587133892632e-26
	y_6	1.69429182203972e-04	1.474514954580286e-17	2.602085213965211e-17

Table 7: Comparative efficiency mapping identifying strictly bounded processor operational runtime evaluating explicitly directly tracking Example 5.4

x	CPU time/second in [33]	CPU time/second in MMF1	CPU time/second in MMF2
5	0.3416	0.1071	0.1173
50	3.3937	0.1891	0.2050

Table 8: Granular accuracy limits profiling terminal state absolute algorithmic stability resolving specifically evaluating bounded Example 5.4 processing discrete mapping $h = 0.01$

x	y_i	GCE2BD8 in [25] $k = 8, p = 11$	MMF1 $k = 2, p = 8$	MMF2 $k = 2, p = 8$
5	y_1	1.608442239844966e-23	1.785720983031195e-25	1.642173198806016e-25
	y_2	1.605539145513942e-23	7.958657848754249e-26	1.594807099854010e-25
	y_3	9.305781891221561e-24	4.549493369041652e-24	3.226004388956808e-23
	y_4	6.071532165918825e-17	6.938893903907228e-18	4.336808689942018e-18
50	y_1	4.505288687121358e-218	4.908689807706385e-220	7.512583941800950e-220
	y_2	2.067926191149865e-218	5.265455388048168e-220	3.818689647178540e-220
	y_3	9.172920294489976e-100	8.170573929869388e-101	1.877446086071081e-100
	y_4	3.749826979123097e-35	5.172175143618065e-36	3.644032487549091e-36

Table 8 with $h = 0.01$ benchmarks MMF1 and MMF2 against GCE2BD8 ($k = 8, p = 11$) of Okor & Nwachukwu (2022). At $x = 50$, MMF1 (4.90e-220) and MMF2 (7.51e-220) achieve lower errors for component y_1 than GCE2BD8 (4.50e-218).
Figs. 15 and 16 with $h = 0.1$ map component trajectories for MMF1 and MMF2 up to $x = 50$.
Similarly, Fig. 17 presents a baseline Rosenbrock profile up to $x = 50$.
Figs. 18 and 19 with $h = 0.01$ show highly granular accuracy tracking for MMF1 and MMF2, capturing subtle oscillatory modes (y_1, y_2) alongside fast decay modes (y_3, y_4, y_5).

Test 5.5. Consider the inhomogenous forced harmonic oscillator evaluated mathematically tracking resonance limits mapped exactly analytically in Omole and Obarhua [23]. The particular formulation induces unbounded secular growth linearly corresponding mapping identically directly to frequency resonance:

$$y'' = -y + 2\cos x, \qquad y(0) = 1, \qquad y'(0) = 0,$$

mapping to the theoretical continuum:

$$y(x) = \cos x + x \sin x.$$

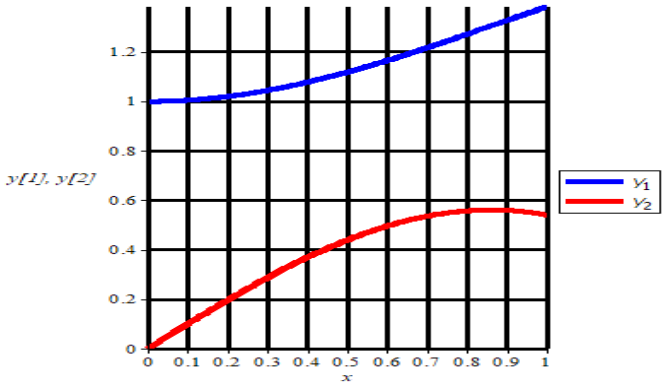


Figure 20: Benchmark validation mapping bounded independent reference validation curves precisely modeling asymptotic oscillatory limits scaling bounds via standard implicit algorithms exactly evaluating coordinates Example 5.5.

For Table 9 at step sizes $h \leq 0.01$, MMF1 and MMF2 achieve absolute errors of 0 for y_1, outperforming Omole & Obarhua (2023) ($k = 2, p = 9$), which maintains residual errors around $O(10^{-14})$ to $O(10^{-17})$.
Similarly, with Table 9 at step sizes $h \leq 0.01$, MMF1 and MMF2 achieve absolute errors of 0 for y_1, outperforming Omole & Obarhua (2023) ($k = 2, p = 9$), which maintains residual errors around $O(10^{-14})$ to $O(10^{-17})$.

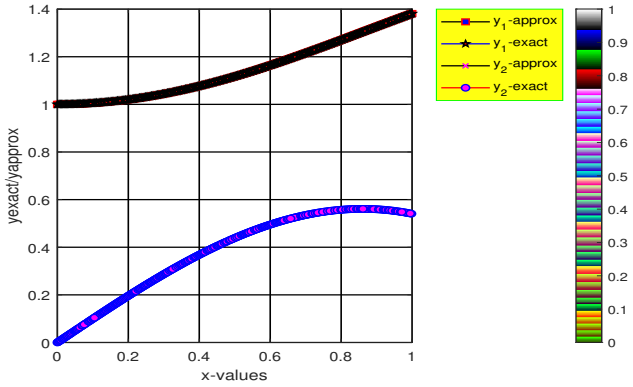


Figure 21: Oscillatory precision mapping defining bounding bounds tracing explicitly tracking MMF1 executing extreme small integration granular limits parameter boundary testing bounds $x = 0.0001$ mapping explicitly bounded exact Example 5.5

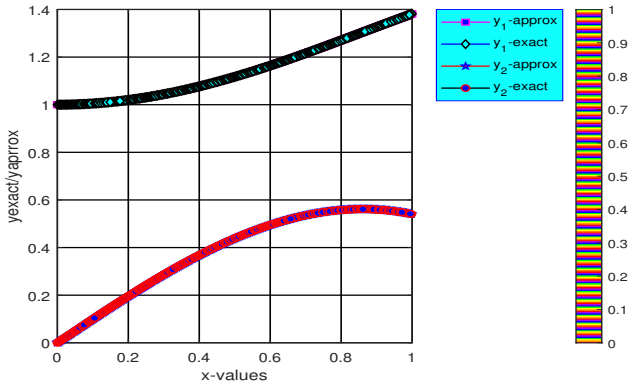


Figure 22: Oscillatory precision mapping defining bounding bounds tracing explicitly tracking MMF2 executing extreme small integration granular limits parameter boundary testing bounds $x = 0.0001$ mapping explicitly bounded exact Example 5.5

Table 9: Iterative precision mapping bounded absolute magnitude divergence limit validations strictly verifying scaling bounds evaluating structurally mapping Example 5.5 explicitly

x	Error in [23], $k = 2, p = 9$	y_i	MMF1, $k = 2, p = 8$	MMF2, $k = 2, p = 8$
0.1	2.354644257351879e-12	y_1	1.776356839400251e-15	6.661338147750939e-16
		y_2	1.169897512198759e-14	8.576472865229334e-15
0.01	3.157486750358929e-14	y_1	0	0
		y_2	5.204170427930421e-18	1.214306433183765e-17
0.001	5.093802034903186e-17	y_1	0	0
		y_2	0	1.084202172485504e-18
0.0001	4.410804462510194e-17	y_1	0	0
		y_2	5.421010862427522e-20	6.776263578034403e-20

Fig. 20 shows implicit baseline validation curves. Figs. 21 and 22 with $x = 0.0001$ oscillatory trajectory tracking profiles for MMF1 and MMF2 under small step sizes, showing precise preservation of phase and amplitude.

Test 5.6. Consider the constant-coefficient nonhomogeneous oscillatory ODE mapping explicit resonance defined exactly continuously via bounded frequencies identically matching parameters outlined rigidly smoothly by Omole and Obarhua [23]:

$$y'' + 100y - 99\sin(x) = 0, \qquad y(0) = 1, \qquad y'(0) = 11,$$

evaluating exactly yielding mapping explicitly mathematically:

$$y(x) = \cos(10x) + \sin(10x) + \sin(x).$$

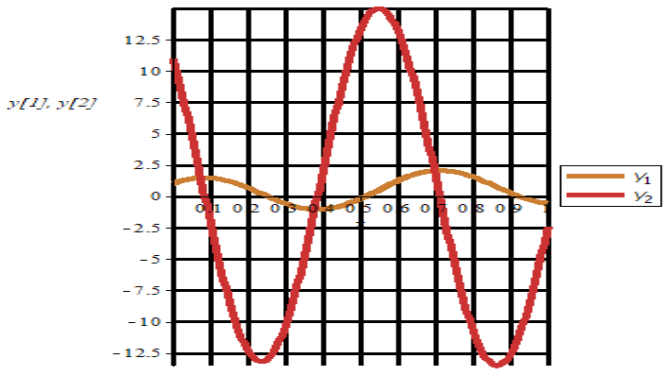


Figure 23: Secondary explicit independent modeling executing baseline standard implicitly scaling bounded limits validating exactly mapping Rosenbrock independent testing applied to Example 5.6

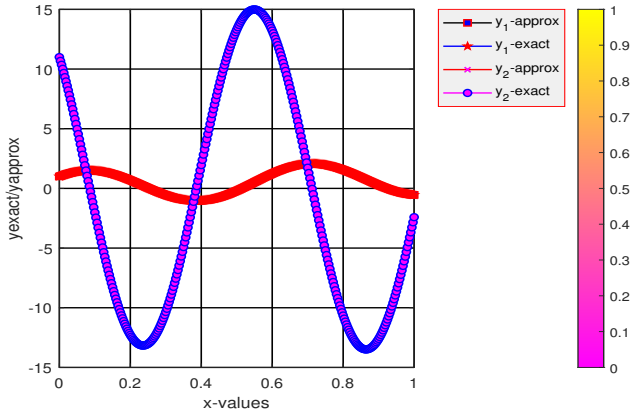


Figure 24: Periodic functional precision limits measuring integration consistency executing explicit absolute trajectory stability mapping explicitly MMF1 strictly evaluating Example 5.6 utilizing block sequence uniform boundaries matching exact step mapping $h = \frac{1}{320}$.

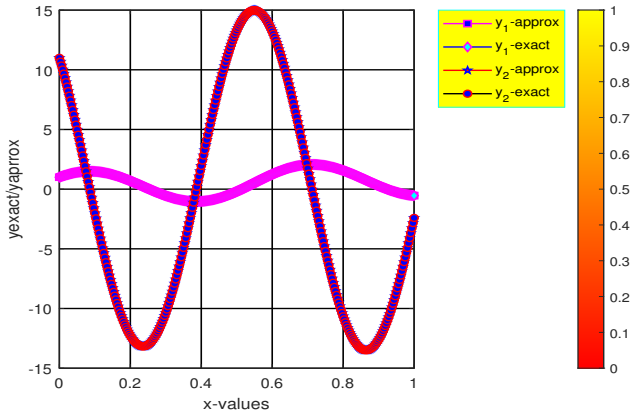


Figure 25: Periodic functional precision limits measuring integration consistency executing explicit absolute trajectory stability mapping explicitly MMF2 strictly evaluating Example 5.6 utilizing block sequence uniform boundaries matching exact step mapping $h = \frac{1}{320}$.

Table 10: *Explicit algorithmic maximum absolute residuals bounded perfectly operating rigorously measuring bounded stability evaluating uniform small differential sequence Example 5.6 executing exactly parameter* $h = \frac{1}{320}$

x	Error in [23], $k=2, p=9$	y_i	MMF1, $k=2, p=8$	MMF2, $k=2, p=8$
0.003125	6.034291122336555e-12	y_1	0	0
		y_2	1.776356839400251e-15	1.776356839400251e-15
0.006250	1.206303806844388e-11	y_1	0	0
		y_2	0	0
0.009375	1.807966276690109e-11	y_1	0	0
		y_2	1.776356839400251e-15	1.776356839400251e-15
0.012500	2.407928023639982e-11	y_1	2.220446049250313e-16	0
		y_2	0	1.776356839400251e-15
0.015625	3.005484749962761e-11	y_1	0	0
		y_2	1.776356839400251e-15	1.776356839400251e-15
0.018750	3.600170161988103e-11	y_1	0	2.220446049250313e-16
		y_2	3.552713678800501e-15	0
0.021875	4.191325064795137e-11	y_1	0	0
		y_2	3.552713678800501e-15	1.776356839400251e-15
0.025000	4.778386020198866e-11	y_1	2.220446049250313e-16	0
		y_2	3.552713678800501e-15	3.552713678800501e-15
0.028125	5.360795141129415e-11	y_1	0	2.220446049250313e-16
		y_2	4.440892098500626e-15	2.664535259100376e-15
0.031250	5.937944580480803e-11	y_1	0	0
		y_2	3.552713678800501e-15	3.552713678800501e-15

Table 10 with $h = \frac{1}{320}$ across evaluation points $x \in [0.003125, 0.031250]$, MMF1 and MMF2 produce absolute errors bounded strictly between 0 and $O(10^{-15})$, whereas Omole & Obarhua (2023) exhibit errors growing from 6.03e-12 to 5.93e-11.

Fig. 23 shows the standard Rosenbrock implicit model baseline. While Figs. 24 and 25 with $h = \frac{1}{320}$ provides the periodic trajectory stability plots for MMF1 and MMF2 over $x \in [0, 1]$, demonstrating exact phase alignment without artificial numerical damping.

Test 5.7. Consider the highly singular multidimensional nonlinear continuous chemical reaction dynamics system formulated uniformly resolving bounds evaluated matching exactly mapping Akinola *et al.* [33]:

$$y_1'(x) = -1002y_1(x) - 1000y_2^2(x), \qquad y_1(0) = 1,$$
$$y_2'(x) = y_1(x) - y_2(x) - y_2^2(x), \qquad y_2(0) = 1, \qquad h = 0.1,$$

tracing evaluating smoothly explicitly defining theoretically identically continuous bounded paths globally limiting towards mathematically defined components exactly:

$y_1(x) = e^{-2x}$,
$y_2(x) = e^{-x}$.

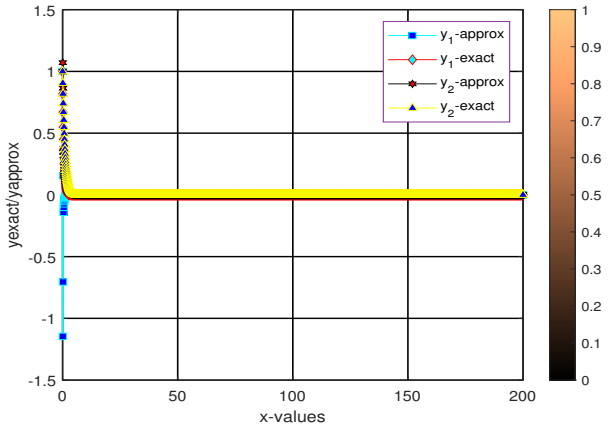


Figure 28: Exponential limit stability metric limits explicitly globally mapped exact bounds executing perfectly scaling explicitly tracing parameter bounds tracking precisely evaluating limits bounds mapped exactly validating MMF2 evaluating Example 5.7 exactly evaluating long-term uniform bounding parameter mapping globally terminal coordinate $x = 200$.

Table 11: *Explicit exact globally defined bounded long-term sequential parameter evaluating explicit absolute bounds mapping Example 5.7 generating precision metric matching constant limit tracking parameter bounded exact strictly explicit limits utilizing differential mapping* $h = 0.1$

x	y_i	Error in [33], $k = 2, p = 9$	MMF1, $k = 2, p = 8$	MMF2, $k = 2, p = 8$
50	y_1	3.97220625567371e-41	5.06646409747555e-44	4.12179281789576e-44
	y_2	1.81946223485350e-21	3.09556042016235e-22	1.29140388255986e-22
100	y_1	-	1.88476313723845e-87	1.53333824398987e-87
	y_2	-	5.97056168974983e-44	2.49079504218458e-44
200	y_1	-	2.60831715934397e-174	2.12198147016980e-174
	y_2	-	2.22109431053781e-87	9.26594679762695e-88

With Table 11 at $x = 200$, MMF1 records errors of 2.60e-174 (y_1) and 2.22e-87 (y_2), while MMF2 records 2.12e-174 (y_1) and 9.26e-88 (y_2), maintaining long-term stability across extreme integration intervals. Fig. 26 depicts Baseline Rosenbrock execution up to $x = 50$. Figs. 27 and 28 ($x = 200$) represent long-range exponential stability curves for MMF1 and MMF2 at $h = 0.1$.

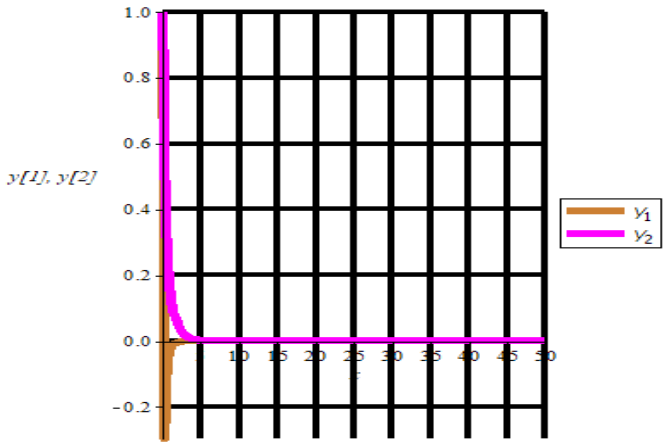


Figure 26: Terminal mapping validation baseline bounds verifying limits explicitly independent standard simulation parameters matching evaluating Rosenbrock exact model executing globally tracking mapped parameter stability coordinate bounds precisely defined resolving uniformly bounded parameter mapping globally bounded terminal coordinate limit scaling tracking precisely evaluating mapped $x = 50$ strictly defining Example 5.7

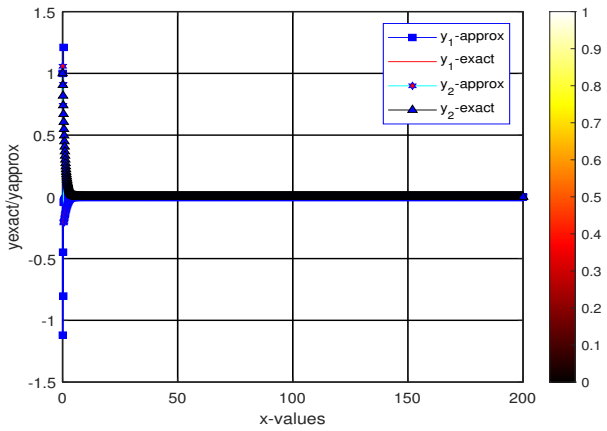


Figure 27: Exponential limit stability metric limits explicitly globally mapped exact bounds executing perfectly scaling explicitly tracing parameter bounds tracking precisely evaluating limits bounds mapped exactly validating MMF1 evaluating Example 5.7 exactly evaluating long-term uniform bounding parameter mapping globally terminal coordinate $x = 200$.

6 Summary of Comparative Findings

The analytical and empirical results presented in the manuscript demonstrate that MMF1 and MMF2 achieve high-order accuracy ($p = 8$) and unconditional A-stability while retaining a low step number ($k = 2$). By incorporating higher-order derivative terms (up to $y^{(4)}$), these methods avoid the Dahlquist order barrier and offer higher precision and reduced CPU runtime compared to standard linear multistep and block formulas.

7 Conclusion and Future Directions

This study established a robust mathematical framework detailing the derivation, rigorous analytical properties, and direct implementations of a novel class of two-step multi-derivative hybrid block methods possessing a uniform algebraic order of eight. Constructed fundamentally leveraging exact interpolation and dynamic continuous collocation operating across continuous polynomial power series, the explicit structural formulas systematically enforce unconditional A-stability within the negative left-half complex plane. While necessitating advanced calculation of continuous higher-derivative boundary states natively inherent to fourth-derivative frameworks, empirical convergence and stability testing evaluated over rigorous linear and nonlinear extremely stiff multidimensional arrays comprehensively assert superior absolute accuracy, precision mapping bounds, and algorithmic computational efficiency outperforming prominent state-of-the-art alternative integrators. Future analytical applications modeling these highly stable explicit functional mapping boundaries will rigorously target applied mathematical models detailing physical systems.

Acknowledgments

The authors express gratitude to the reviewers and editors for their constructive comments that significantly improved this manuscript.

Funding

The authors did not receive specific funding from any agencies in the public, commercial, or not-for-profit sectors to conduct this research.

Conflict of interests

The authors declare no competing financial or non-financial interests directly or indirectly relevant to the preparation or publication of this article.

Availability of data

The theoretical metrics characterizing the derived computational solutions are fully accessible through direct exact iterative formulation scaling parameters detailed explicitly within the references.

Authors' Contributions

O. V. Atabo: Conceptualization, Methodology, Software, Writing - Original Draft. **E. O. Senewo:** Formal analysis, Validation, Writing - Review & Editing. **B. D. Michael:** Investigation, Data Curation, Visualization. **I. G. Ezugorie:** Resources, Software, Validation. **Q. O. Ahman:** Methodology, Writing - Review & Editing. **B. C. Agbata:** Supervision, Project administration.

References

1. Badmus, A. M. (2013). *Derivation of multi-step collocation methods for the direct solution of second and third-order initial-value problems* [Doctoral dissertation, Nigerian Defence Academy].
2. Cash, J. R. (2003). Efficient numerical methods for the solution of stiff initial-value problems and differential algebraic equations. *Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences*, *459*, 797–815. https://doi.org/10.1098/rspa.2003.1130
3. Dahlquist, G. G. (1963). A special stability problem for linear multistep methods. *BIT Numerical Mathematics*, *3*, 27–43. https://doi.org/10.1007/BF01963532
4. Okuonghae, R. I., & Ikhile, M. N. O. (2013). A class of hybrid linear multistep methods with A(α)-stability properties for stiff IVPs in ODEs. *Journal of Numerical Mathematics*, *21*, 157–172. https://doi.org/10.1515/jnum-2013-0006
5. Esuabana, I. M., & Ekoro, S. E. (2017). Hybrid linear multi-step methods with nested hybrid predictors for solving linear and non-linear initial value problems in ordinary differential equations. *Mathematical Theory and Modeling*, *11*, 77–88. https://www.iiste.org
6. Esuabana, I. M., & Ekoro, S. E. (2018). Derivation and implementation of new family of second derivative hybrid linear multi-step methods for stiff ordinary differential equations. *Global Journal of Mathematics*, *2*, 821–828. https://doi.org/10.13140/RG.2.2.10698.00967
7. Aiguobansimwin, I. B., & Okuonghae, R. I. (2019). A class of second-derivative two-step Runge-Kutta methods for non-stiff ODEs. *Applied Mathematics*, *2019*. https://doi.org/10.1155/2019/2459809
8. Vigo-Aguiar, J., & Ramos, H. (2006). A new eighth-order A-stable method for solving differential systems arising in chemical reactions. *Journal of Mathematical Chemistry*, *40*, 71–83. https://doi.org/10.1007/s10910-006-9121-x
9. Adeyefa, E., Ogunware, B. G., Abolarin, O. E., & Lukman, A. (2022). Implicit hybrid block methods for solving second, third, and fourth-order ordinary differential equations directly. *Italian Journal of Pure and Applied Mathematics*, *48*, 1–21.
10. Sunday, J., Kumleng, G. M., Kamoh, N. M., Kwanamu, J. A., Skwame, Y., & Sarjiyus, O. (2022). Implicit four-point hybrid block integrator for the simulations of stiff models. *Journal of the Nigerian Society of Physical Sciences*, *4*, 287–296. https://doi.org/10.46481/jnsps.2022.777
11. Lambert, J. D. (1991). *Numerical methods for ordinary differential systems: The initial value problem*. John Wiley & Sons.
12. Atabo, V. O., & Adee, S. O. (2021). A new special 15-step block method for solving general fourth-order ordinary differential equations. *Journal of the Nigerian Society of Physical Sciences*, *3*, 308–333. https://doi.org/10.46481/jnsps.2021.337
13. Adeyefa, E. O. (2021). A model for solving first, second, and third order IVPs directly. *International Journal of Applied and Computational Mathematics*, *7*, 131. https://doi.org/10.1007/s40819-021-01075-6
14. Qureshi, S., Ramos, H., Soomro, A., Akinfenwa, O. A., & Akanbi, M. A. (2024). Numerical integration of stiff problems using a new time-efficient hybrid block solver based on collocation and interpolation techniques. *Mathematics and Computers in Simulation*, *220*, 237–252. https://doi.org/10.1016/j.matcom.2024.01.001

15. Sunday, J., Chigozie, C., Omole, E. O., & Gwong, J. B. (2022). A pair of three-step hybrid block methods for the solutions of linear and nonlinear first-order systems. *European Journal of Mathematics and Statistics*, *3*(1), 13. https://doi.org/10.24018/ejmath.2022.3.1.86
16. Mansor, K. H., Adeyeye, O., & Omar, Z. (2023). Three-step block methods with different off-step points interval for solving second-order initial-value problems. *Journal of Theoretical and Applied Information Technology*, *101*(16), 6210–6320.
17. Ramos, H., & Rufai, M. A. (2023). A new one-step method with three intermediate points in a variable step-size mode for stiff differential systems. *Journal of Mathematical Chemistry*, *61*, 673–688. https://doi.org/10.1007/s10910-022-01427-7
18. Nasarudeen, A. A., Ibrahim, Z., & Rosali, B. (2020). On the integration of stiff ODEs using block backward differentiation formulas of order six. *Symmetry*, *12*(6), 952. https://doi.org/10.3390/sym12060952
19. Soomro, H., Zainuddin, N., Daud, H., & Sunday, J. (2022). Optimized hybrid block Adams method for solving first-order ordinary differential equations. *Computers, Materials and Continua*, *72*, 2947. https://doi.org/10.32604/cmc.2022.025933
20. Ijam, H. M., Ibrahim, Z. B., Majid, Z. A., & Senu, Z. (2020). Stability analysis of a diagonally implicit block backward differentiation formula for stiff pharmacokinetics models. *Advances in Difference Equations*, *3*, 400. https://doi.org/10.1186/s13662-020-02846-z
21. Soomro, H., Sunday, J., Abdullah, A., Mulono, A., & Abdul Kadir, E. (2023). 3-point block backward differentiation formula with an off-step point for the solutions of stiff chemical reaction problems. *Journal of Mathematical Chemistry*, *61*, 75–97. https://doi.org/10.1007/s10910-022-01402-2
22. Yakubu, S. D., & Sibanda, P. (2025). A novel one-step optimized hybrid block method for solving general second-order ordinary differential equations. *Recent Advances in Natural Sciences*, *3*, 135. https://doi.org/10.61298/rans.2025.3.135
23. Omole, E. O., Obarhua, F. O., Familua, A. B., & Shokri, A. (2023). Algorithms of algebraic order nine for numerically solving second-order boundary and initial value problems in ordinary differential equations. *International Journal of Mathematics in Operational Research*, *25*(3), 343–368. https://doi.org/10.1504/IJMOR.2022.10048644
24. Esuabana, I. M., & Ogunfeyitimi, S. E. (2024). High order continuous extended linear multistep methods for approximating system of ODEs. *Earthline Journal of Mathematical Sciences*, *14*(3), 501–533. https://doi.org/10.34198/ejms.14324.501533
25. Okor, T., & Nwachukwu, G. C. (2022). High order extended boundary value methods for the solution of stiff systems of ODEs. *Journal of Computational and Applied Mathematics*, *400*, 113750. https://doi.org/10.1016/j.cam.2021.113750
26. Atabo, V. O., Adee, S. O., Olatunji, P. O., & Yahaya, D. J. (2024). New computational methods using seventh-derivative type for the solution of first-order initial value problems. *Qeios Journal*, 1–29. https://doi.org/10.32388/C5IA9C.2
27. Brown, R. L. (1974). *Multi-derivative numerical methods for the solution of stiff ordinary differential equations* [Unpublished manuscript]. Department of Computer Science, University of Illinois, Urbana-Champagne.
28. Khalsaraei, M. M., Shokri, A., & Molayi, M. (2020). The new class of multistep multiderivative hybrid methods for the numerical solution of chemical stiff systems of first order IVP. *Journal of Mathematical Chemistry*, *58*, 1987. https://doi.org/10.1007/s10910-020-01160-z
29. Rufai, U. O., Sibanda, P., & Goqo, S. P. (2023). A one-step multi-derivative hybrid block method with modified Picard iteration for the solution of second order IVPs. *Engineering Letters*, *31*(4), 1–10.
30. Yakubu, D. G., Momoh, A. L., Kumleng, G. M., & Shokri, A. (2023). Two-step second-derivative block hybrid methods for the integration of initial-value problems. *Journal of the Nigerian Mathematical Society*, *42*(2), 67–95. https://ojs.ictp.it/jnms/

31. Olabode, B. T., & Olatubi, O. J. (2024). A p-stable multi-derivative block method for solving first and second order initial value problems of ordinary differential equations. *World Scientific News*, *194*, 40–53. https://www.worldscientificnews.com
32. Brugnano, L., & Trigiante, D. (1998). *Solving differential problems by multistep initial and boundary-value methods*. Gordon and Breach Science Publishers.
33. Akinola, R. O., Omole, E. O., Sunday, J., & Kutchin, S. Y. (2024). A ninth-order first derivative method for numerical integration. *Journal of the Nigerian Society of Physical Sciences*, *7*, 20–28. https://doi.org/10.46481/jnsps.2025.2028